AWS VPC PEERING

Secure Private Communication Between Two Virtual Private Clouds

Project / Internship	AWS Cloud Computing Internship Project
Organization	EWB Courses
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Region	Asia Pacific (Mumbai) — ap-south-1
Report Date	20 October 2026

Project summary

This project configures two Amazon Virtual Private Clouds (VPCs) connected by a VPC peering connection. A Bastion EC2 instance in the public subnet of VPC A provides an SSH administration path to the private application instances. Route tables and security groups control private traffic between the VPCs.

1. Objective

The objective is to establish and verify private network communication between two VPCs using VPC Peering, route tables, security groups, and EC2 instances. A Bastion host is used as the SSH entry point for administering private instances.

2. Network Architecture and IP Plan

The implementation follows the EWB project architecture supplied for this assignment.

Component	Name / Address	Purpose
VPC A	ewb-vpc-a — 10.10.0.0/16	Hosts Bastion and App-A
VPC B	ewb-vpc-b — 10.20.0.0/16	Hosts App-B; no internet gateway
Public subnet A	ewb-a-public — 10.10.1.0/24	Bastion subnet
Private subnet A	ewb-a-private — 10.10.2.0/24	App-A subnet
Private subnet B	ewb-b-private — 10.20.1.0/24	App-B subnet
Bastion	10.10.1.10	SSH entry point
App-A	10.10.2.10	Private workload in VPC A

App-B	10.20.1.10	Private workload in VPC B
AWS Region	ap-south-1 (Mumbai)	Region used for resources

3. AWS Resources and Configuration

VPCs and subnets: Two non-overlapping VPC CIDR blocks were used. VPC A contains one public subnet and one private subnet; VPC B contains one private subnet.

Internet access: An Internet Gateway is attached to VPC A for the Bastion's public access. VPC B is designed without an Internet Gateway.

VPC Peering: The connection named ewb-a-b-peer was created between VPC A and VPC B and was observed in the Active state.

Routing: The public route table for VPC A includes a default route to the Internet Gateway and a route to VPC B through the peering connection. The private route tables include routes to the opposite VPC CIDR through the peering connection.

Security groups: ewb-sg-bastion controls SSH access to the Bastion. ewb-sg-app-a and ewb-sg-app-b restrict SSH and ICMP access to the private instances. The controlled test temporarily removed the ICMP permission for traffic from App-A to App-B and then restored it.

Compute: Three Amazon Linux 2023 EC2 instances were used: Bastion, App-A, and App-B.

4. Administration Path

SSH ProxyJump was configured on the Windows laptop so that App-A and App-B could be reached through the Bastion. Direct SSH from the laptop to a private IP is not expected to work because the application instances have private IP addresses only.

Administration paths tested:

• Laptop → Bastion → App-A (10.10.2.10)

• Laptop → Bastion → App-B (10.20.1.10)

5. Testing and Results

The following results are based on the tests performed during the project. Confirm the post-restoration test before treating the recovery result as final.

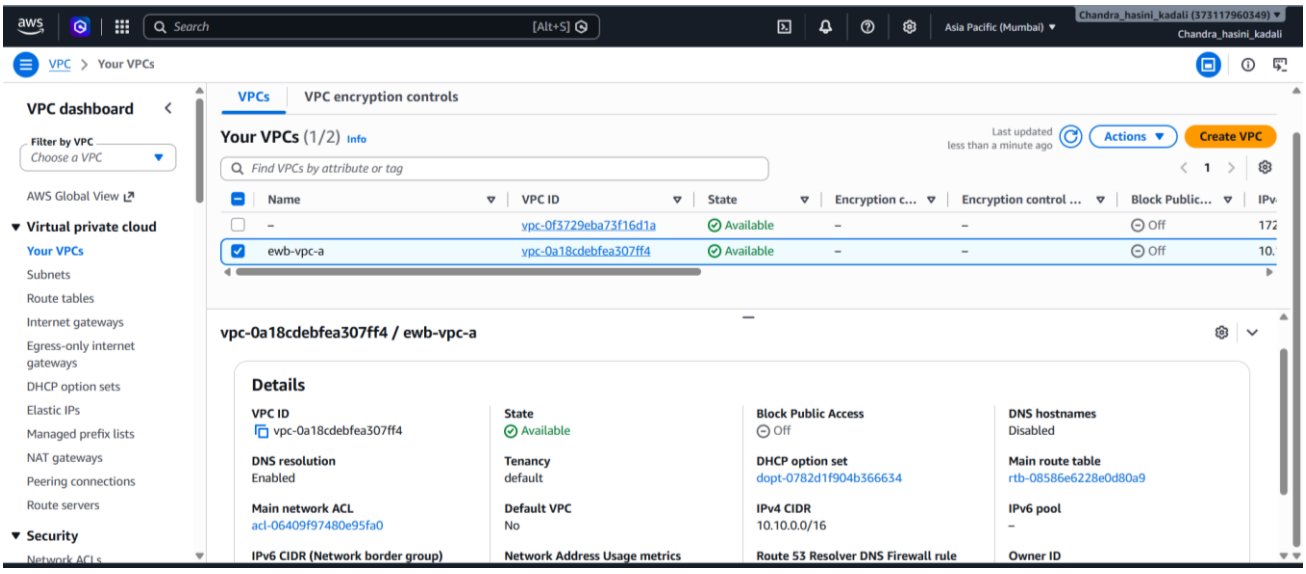
Test	Observed result	Interpretation
SSH from laptop to Bastion	Successful	Public entry point reachable using the configured key
SSH ProxyJump to App-A	Successful	Private instance reachable through Bastion
SSH ProxyJump to App-B	Successful	Private instance reachable through Bastion
App-A → App-B ICMP	4 sent, 4 received; 0% loss	Private communication succeeded before controlled failure
App-B → App-A ICMP	4 sent, 4 received; 0% loss	Reverse private communication succeeded
Controlled failure	100% packet loss after removing the ICMP rule	Security group rule controlled the tested ICMP traffic

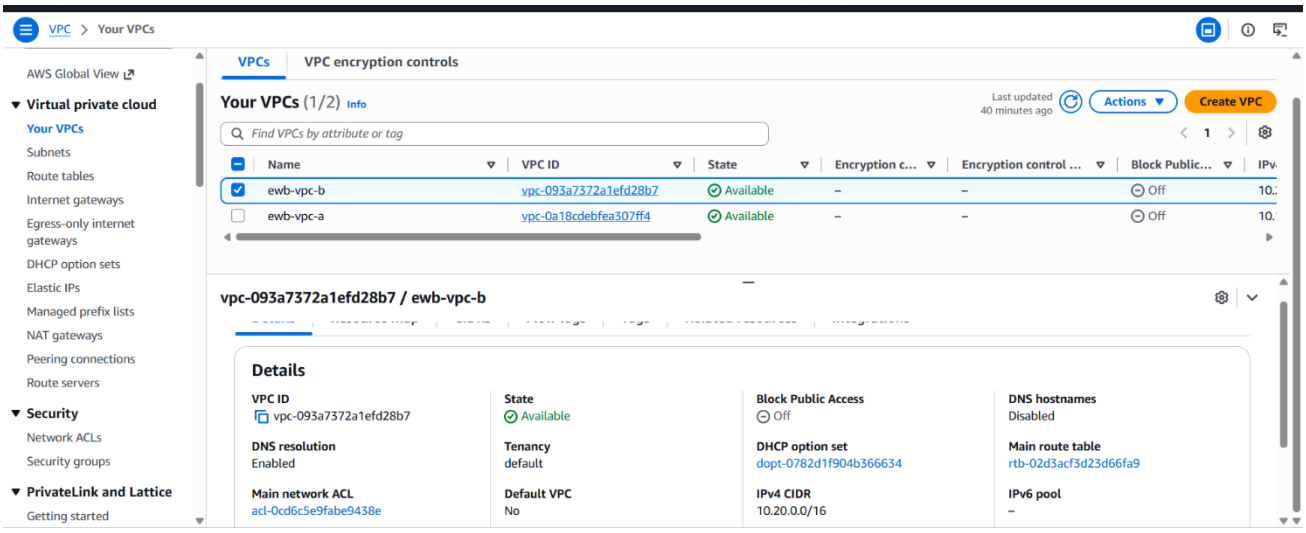
Rule restoration	Rule restored; final ping confirmation to be recorded	Rerun ping and record 0% loss after restoration

6. Screenshot Checklist

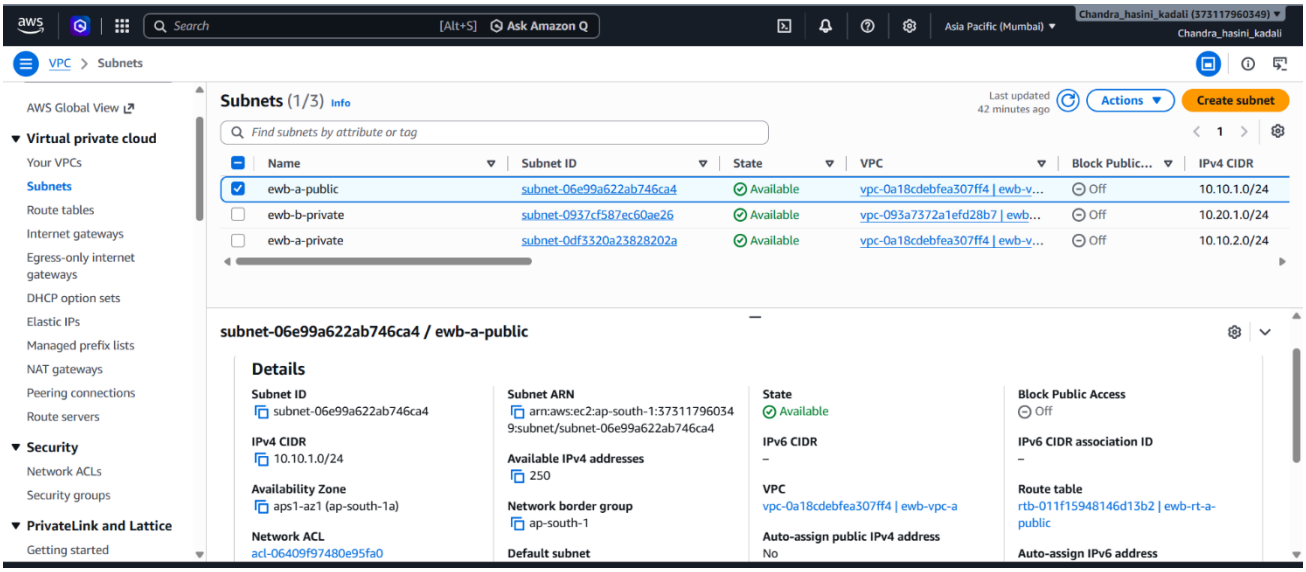
No.	Screenshot evidence	Done
1	VPC A and VPC B details/CIDR blocks	■
2	Subnet details and CIDR blocks	■
3	EC2 instances with private IP addresses	■
4	VPC peering connection showing Active	■
5	Route table entries pointing to the peering connection	■
6A	SSH ProxyJump connection to App-A	■
6B	SSH ProxyJump connection to App-B	■
7	Successful App-A → App-B ping	■
8	Successful App-B → App-A ping	■
9	Controlled failure showing 100% packet loss	■
10	Successful ping after restoring the ICMP rule	■

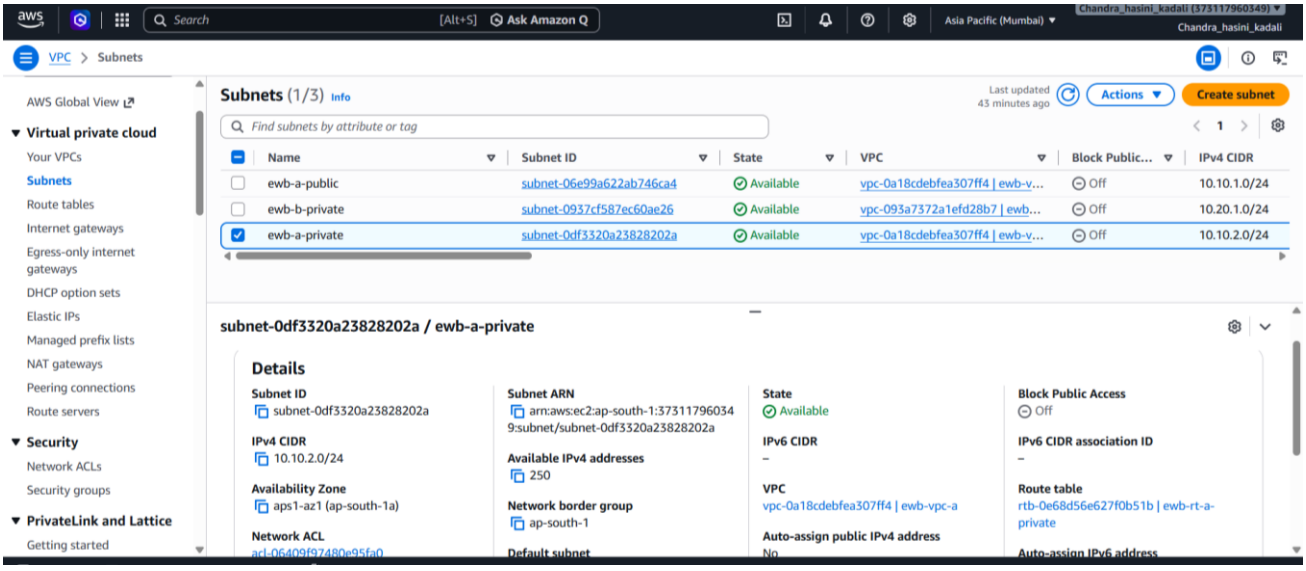
Screenshot 1: VPC Configuration

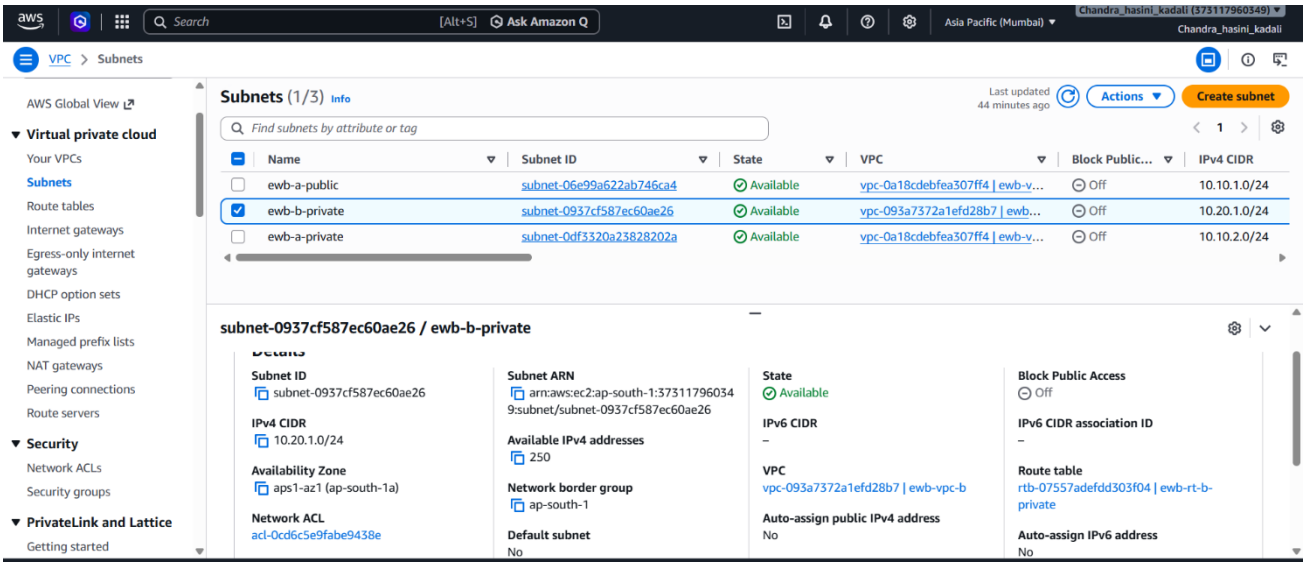




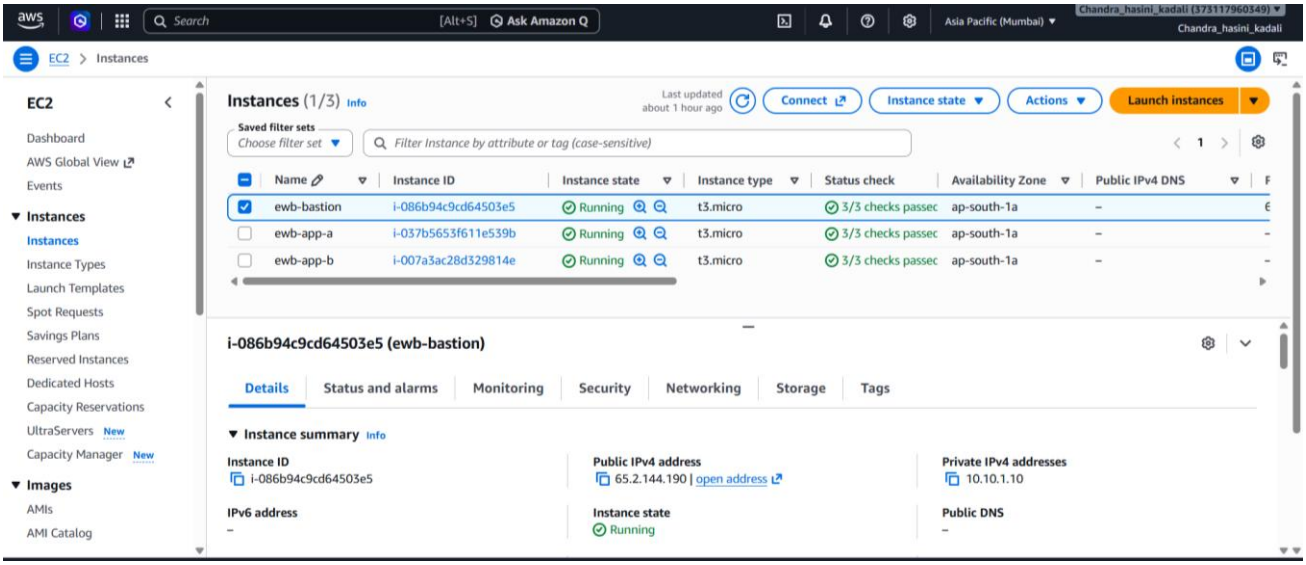
Screenshot 2: Subnet Configuration

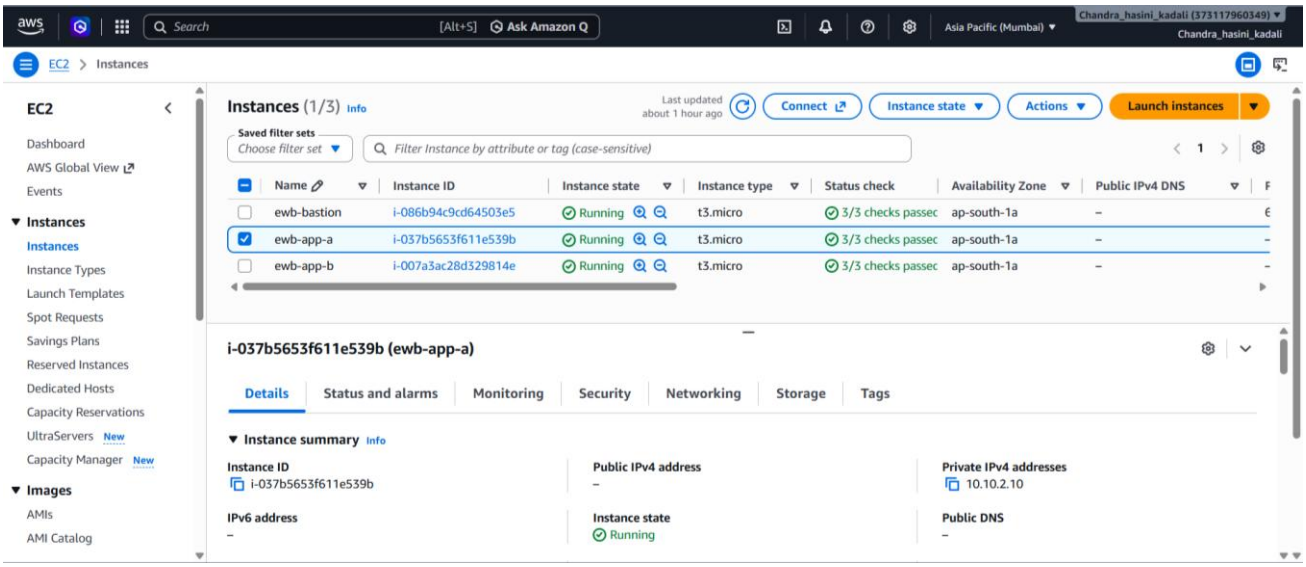


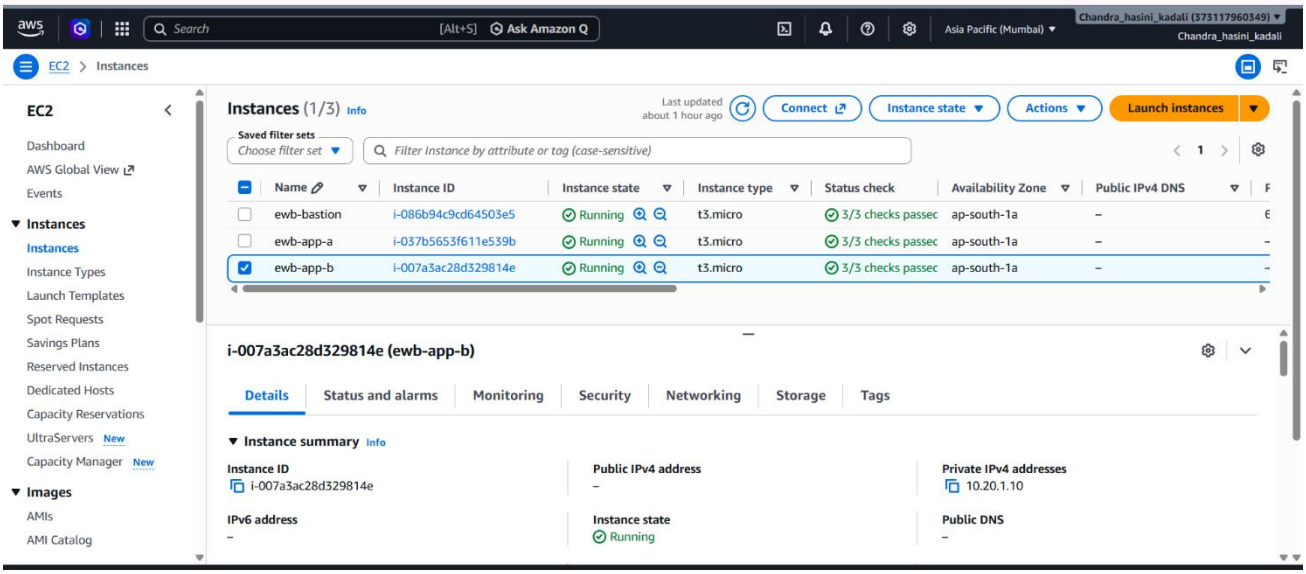




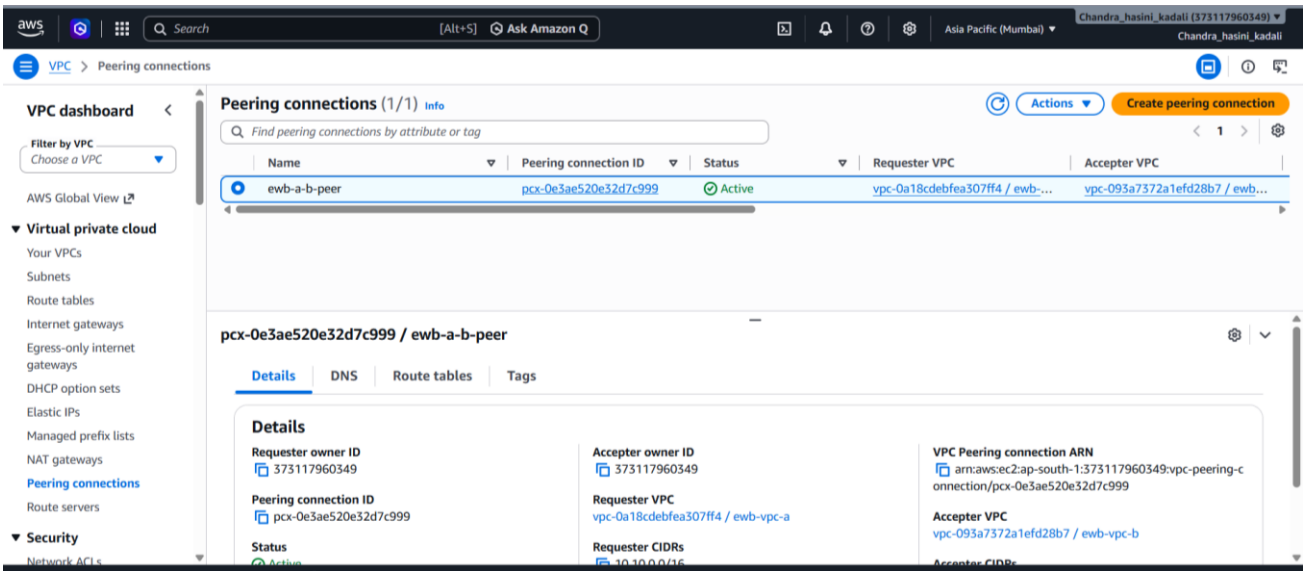
Screenshot 3: EC2 Instances



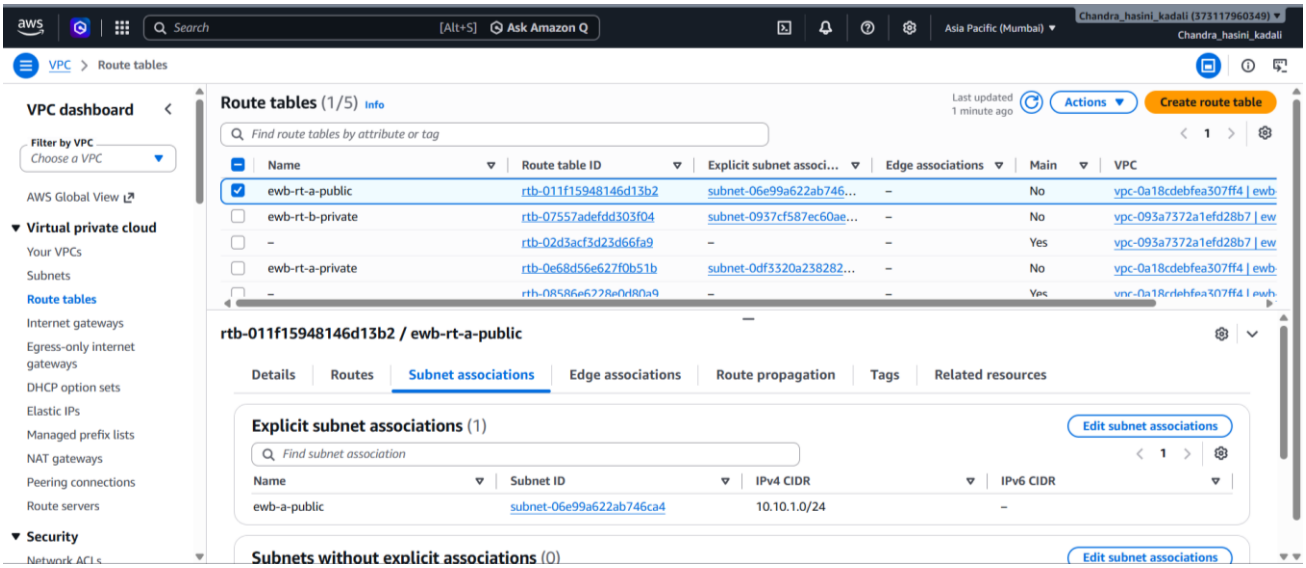


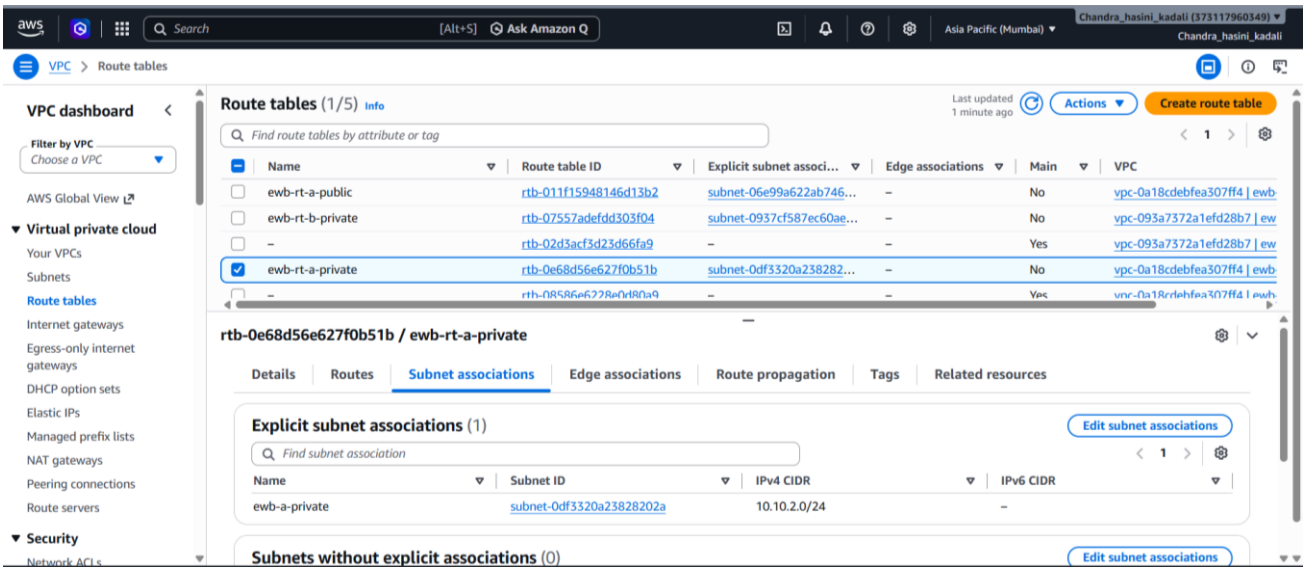


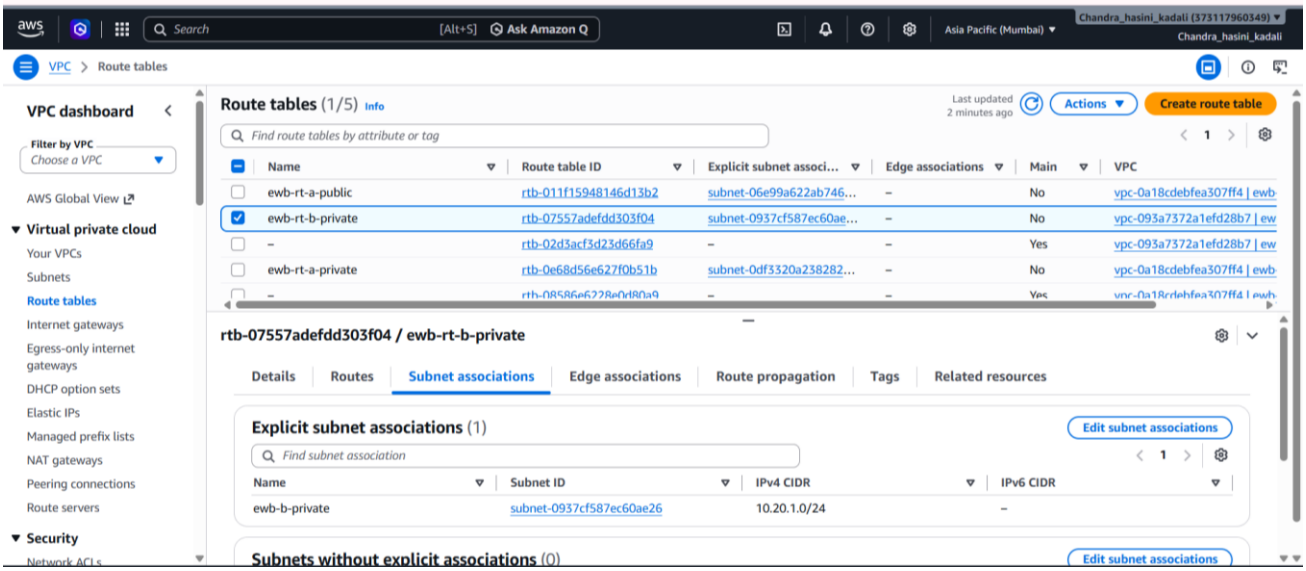
Screenshot 4: VPC Peering Connection



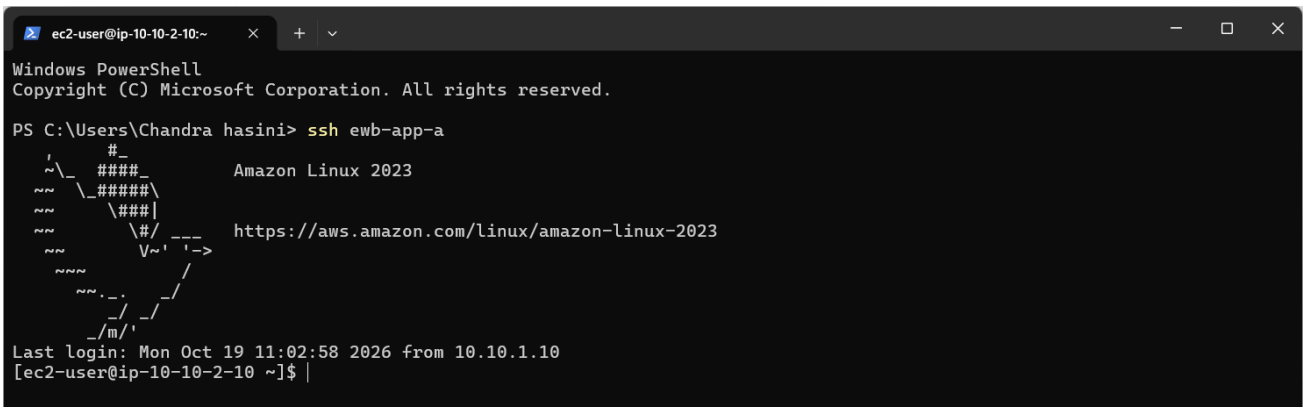
Screenshot 5: Route Tables







Screenshot 6: SSH Connectivity



```
PS C:\Users\Chandra hasini> ssh ewb-app-b
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 11:29:32 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$
```

Screenshot 7: Successful Connectivity Test

```
ec2-user@ip-10-20-1-10:~
    ProxyJump ewb-bastion
PS C:\Users\Chandra hasini> ssh ewb-app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:nV0W/slRgXqlU0RptUKoXvHo12HysMCXgLK9z+xBODQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.456 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.224 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.309 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.199 ms

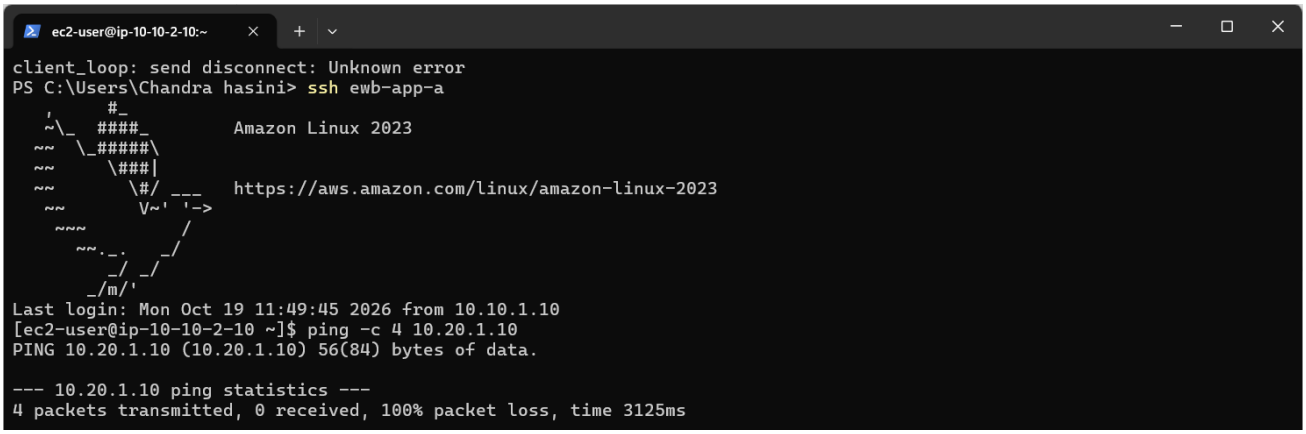
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3130ms
rtt min/avg/max/mdev = 0.199/0.297/0.456/0.100 ms
```

Screenshot 8: Reverse Connectivity Test

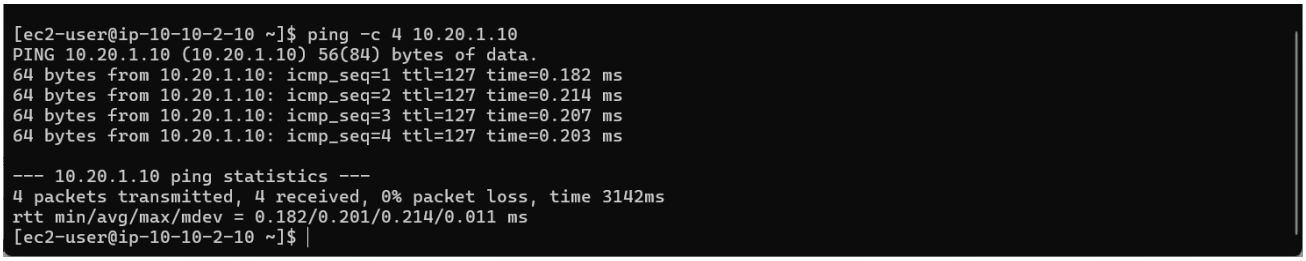
```
ec2-user@ip-10-20-1-10:~
PS C:\Users\Chandra hasini> ssh ec2-user@10.20.1.10
ssh: connect to host 10.20.1.10 port 22: Connection timed out
PS C:\Users\Chandra hasini> notepad "$env:USERPROFILE\.ssh\config"
PS C:\Users\Chandra hasini> ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:KuyDdY7qe0cIj0XbRYTudsT5Bu9tO3WvYT7Lhu4IcYA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.325 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.326 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.314 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.323 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3132ms
rtt min/avg/max/mdev = 0.314/0.322/0.326/0.004 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Screenshot 9: Controlled Failure Test



Screenshot 10: Restored Connectivity



6. Conclusion

The configured VPC peering connection enabled App-A and App-B to communicate over their private IP addresses. Successful ICMP tests in both directions, SSH ProxyJump access through the Bastion, and the controlled failure test demonstrated the roles of routing and security group rules in the network design. The final recovery test should be documented after confirming connectivity following rule restoration.